

Enhanced Multidimensional Time Series Forecasting via Pre-trained Temporal Encoders and GNNs

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Abstract—Accurately forecasting multivariate time series requires capturing both temporal patterns within individual series and relational dependencies across entities. To address this challenge, this paper proposes a unified multimodal forecasting framework from temporal, spatial, and textual perspectives. Specifically, we employ temporal encoders, including pre-trained temporal models and neural networks trained from scratch, to extract short- and long-term temporal features. We further construct a graph-based representation and apply a Graph Neural Network (GNN) to model inter-series dependencies. In addition, cryptocurrency-related news text is incorporated to enrich the feature space. The extracted temporal, graph, and textual features are fused by a Multilayer Perceptron (MLP) for final prediction. Experiments on real-world cryptocurrency market data show that jointly using temporal, relational, and news features improves prediction performance, while the effectiveness of news features depends on data quality, relevance, and preprocessing.

Index Terms—Multivariate time series forecasting, cryptocurrency forecasting, pre-trained temporal encoders, graph neural networks, multimodal feature fusion.

I. INTRODUCTION

Many real-world applications involve forecasting tasks over multiple correlated time series, where individual sequences exhibit complex temporal dynamics while simultaneously interacting with each other through implicit or explicit dependency structures. Effectively modeling such systems requires not only capturing temporal patterns within each sequence, but also characterizing cross-series relationships and incorporating external contextual information. These challenges are common in a wide range of domains, including finance, transportation, energy systems, and social platforms [1], [2], [3], [4]. For example, in urban traffic flow prediction, the flow of each road is influenced not only by its own historical conditions, but also by the congestion of adjacent roads. Similarly, in financial markets, price fluctuations of different stocks contain their own temporal patterns and are influenced by multiple cross-influences of industry relationships and market sentiment.

From a data modeling perspective, multivariate time series typically exhibit non-stationarity, structural heterogeneity, and dynamic dependency patterns [5]. For example, in price forecasting tasks in the financial field, external factors such as policy changes, macroeconomic events, or collective behavioral responses further exacerbate the complexity of the temporal evolution of these series [6], [7]. Consequently, the

relationships between different time series become unstable over time, lacking a persistent long-term equilibrium structure and exhibiting varying degrees of independence or weak coupling [8]. These characteristics pose significant challenges to accurate prediction and representation learning of multivariate time systems. In this work, we take the financial time series as an example to discuss how to improve the predictive performance of multivariate time series using the latest technologies.

Existing studies have explored such problems from multiple perspectives, including temporal dynamics, dependency modeling, and external influence analysis. Traditional approaches often rely on econometric or statistical models, such as GARCH-type models [9], vector autoregression (VAR) models [10], and dynamic conditional correlation analysis [11], to characterize correlations and volatility transmission. Although these methods provide interpretability, their representational capacity is limited when handling complex nonlinear temporal patterns and high-dimensional interactions. With the rapid development of deep learning, neural network-based time series models have become increasingly popular. Early studies primarily adopted classical temporal neural networks (TNNs), such as long short-term memory (LSTM) networks [12] and temporal convolutional networks (TCNs) [13]. More recently, large transformer-style and mixer-based architectures, such as the pre-trained TimesFM [14], have been developed to leverage knowledge from large time series datasets, enabling better capture of long-range dependencies and improving generalization. In parallel, incorporating external unstructured data, such as textual information from news or social media, further enhances predictive performance by providing complementary contextual cues [15]. On the other hand, graph neural networks (GNNs) have been widely adopted to represent relational structures among entities, enabling the modeling of dependency propagation, interaction strength, and collective dynamics [16], [17], [18].

In this work, we propose a Multimodal Multivariate Forecasting Framework (MMPF) that jointly models temporal dynamics, inter-series relationships, and external textual information within a unified architecture. Specifically, temporal neural networks, including LSTM, TCN, and recent pre-trained mixer-style models such as WPMixer [19] and TimeMixer [20] (MLP-based), as well as TimeXer [21] (Transformer-based), trained on multivariate time series datasets including ETTh1, ETTh2, ETTm1, ETTm2, Weather, Electricity, and Traffic,

This work was supported by the National Natural Science Foundation of China under Grant No. 62471309. Xiaoxiao Wu is the corresponding author.

are employed to extract multi-scale temporal features from individual time series. Meanwhile, we also extract textual and semantic representations using a BERT-based model [22] to capture external contextual signals. Graph convolutional networks (GCNs) [23] are utilized to model dynamic relational structures among different entities and to learn patterns of spatial dependency across the multivariate system. The extracted multimodal features are then fused to produce final forecasting results. We evaluate the proposed framework using cryptocurrency price data. Our numerical results demonstrate that fusing this multimodal information can help predict cryptocurrency trends and better understand the relationships between different cryptocurrencies. It is worth highlighting that the proposed method is not limited to cryptocurrency forecasting and can be readily extended to other multivariate time series prediction tasks.

II. METHODOLOGY

A. Problem Formulation

Suppose that there are N cryptocurrencies, indexed by the node set $\mathcal{V} = \{1, \dots, N\}$, and observed over T time steps $t = 1, \dots, T$. Denote the closing price of the currency $i \in \mathcal{V}$ at time t by $x_{i,t} > 0$, and we also define the logarithmic rate of return: $r_{i,t} = \log x_{i,t} - \log x_{i,t-1}$. Stacking across currencies and time produces the price matrix $\mathbf{X} \in \mathbb{R}^{N \times T}$ (or the return matrix $\mathbf{R} \in \mathbb{R}^{N \times T}$). We consider a sliding look-back window of length L and a forecasting horizon $h \geq 1$. As shown in Fig. 1, each reference time t ($L \leq t \leq T - h$), the model observes the window $\mathbf{X}_{t-L+1:t}$ and predicts future prices over the horizon $\mathbf{X}_{t+1:t+h}$. In addition to cryptocurrency prices, we also collect news text related to cryptocurrencies. Let $\mathcal{Z}_{i,\tau} = \{\mathbf{z}_{i,\tau}^{(k)}\}_{k=1}^{K_{i,\tau}}$ denote the set of news items related to currency i at time τ . Each news item $\mathbf{z}_{i,\tau}^{(k)}$ is later processed through a language model (e.g., BERT) to extract a multi-component embedding. We also have a cryptocurrency relationship graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$, with the adjacency matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$. Herein $\mathcal{G}$ may be static or change over time $\{\mathcal{G}_t\}_{t=1}^T$ with the adjacency matrix $\{\mathbf{A}_t\}$. We may construct the adjacency matrix based on price correlations or Pearson correlation coefficients between currencies. Fig. 1 illustrates the framework of the proposed model.

B. The Proposed Multimodal Multi-Currency Forecasting Framework (MMPF)

Our framework extracts three multimodal features from time series, social media text, and multi-cryptocurrency social spatial relationships to help predict price trends. Following Fig. 1, we introduce the proposed framework as follows.

1) Temporal encoder (for price series): For each cryptocurrency i , a temporal encoder f_T extracts a fixed-dimensional representation from the look-back window:

$$\mathbf{F}_{i,t}^T = f_T(\mathbf{x}_{i,t-L+1:t}) \in \mathbb{R}^{d_T}, \quad (1)$$

where $\mathbf{x}_{i,t-L+1:t}$ collect from the i -th row of $\mathbf{X}_{t-L+1:t}$, denoting the prices in the window for the cryptocurrency i . Such temporal encoders are expected to capture sequential

patterns and trends in currency price movements, enabling the model to leverage historical market dynamics. The choice of encoder depends on the temporal dependencies to be captured: TCN and LSTM are trained from scratch to model local patterns and long-term dependencies, respectively, while other pre-trained encoders, such as Transformer-based models, are used to capture attention-driven temporal relationships. In the future, we may also use the inner encoder of large time series-based models (such as TimesFM [14]) as a powerful temporal encoder.

2) Text encoder (for news): To extract the text features, we embed each news item $\mathbf{z}_{i,\tau}^{(k)}$ via a language model (e.g., BERT), to form a news representation:

$$\mathbf{e}_{i,\tau}^{(k)} = [\mathbf{e}^{\text{title}}, \mathbf{e}^{\text{subtitle}}, \mathbf{e}^{\text{body}}, \mathbf{s}, \mathbf{f}_{\text{stat}}, \mathbf{f}_{\text{sent}}] \in \mathbb{R}^{d_e}, \quad (2)$$

where $\mathbf{e}^{\text{title}}, \mathbf{e}^{\text{subtitle}}, \mathbf{e}^{\text{body}} \in \mathbb{R}^{768}$ are the embeddings of the title, subtitle, and main body of the news item, $\mathbf{s} \in \mathbb{R}^3$ contains pairwise cosine similarities between the embeddings of the title, subtitle, and body, $\mathbf{f}_{\text{stat}} \in \mathbb{R}^6$ includes statistical features, and $\mathbf{f}_{\text{sent}} \in \mathbb{R}^2$ represents sentiment values, yielding $d_e = 768 \times 3 + 3 + 6 + 2 = 2315$.

Then, for each currency i at time t , we apply weighted temporal pooling to aggregate news items within a valid window of length W , e.g., news from the previous W days:

$$\mathbf{F}_{i,t}^{\text{news}} = \frac{\sum_{\tau=1}^W \sum_{k=1}^{K_{i,t-\tau+1}} w_{\tau} \mathbf{e}_{i,t-\tau+1}^{(k)}}{\sum_{\tau=1}^W K_{i,t-\tau+1} w_{\tau}} \in \mathbb{R}^{d_{\text{news}}}, \quad (3)$$

where the linear decay weight is defined as $w_{\tau} = 1 - \frac{\tau-1}{W}$, indicating that news farther from the reference time has a smaller impact on the price in the next period. If no news item is available within the valid window, we set $\mathbf{F}_{i,t}^{\text{news}}$ to a zero vector to avoid division by zero.

3) Fusion of temporal and news features: Next, we fuse price and news features using an MLP-based fusion model. Specifically, the temporal representation and the news representation are first concatenated and then projected into a unified latent space:

$$\mathbf{H}_{i,t} = \phi_{\text{MLP}}([\mathbf{F}_{i,t}^T; \mathbf{F}_{i,t}^{\text{news}}]) \in \mathbb{R}^{d_H}, \quad (4)$$

where $[\cdot; \cdot]$ denotes feature concatenation and ϕ_{MLP} is a multilayer perceptron used to fuse price and news representations. By stacking the fused representations of all cryptocurrencies, we obtain $\mathbf{H}_t \in \mathbb{R}^{N \times d_H}$. This fusion module projects heterogeneous temporal and textual features into a unified latent space for subsequent graph-based modeling.

4) Graph fusion (for multicurrency social network): After obtaining the fused features $\mathbf{H}_t$ at reference time t for all cryptocurrencies, we further enhance these representations by extracting the relational structure between different currencies through a graph convolutional network (GCN). Let $\hat{\mathbf{A}} = \mathbf{A} + \mathbf{I}$ and $\tilde{\mathbf{A}} = \hat{\mathbf{D}}^{-1/2} \hat{\mathbf{A}} \hat{\mathbf{D}}^{-1/2}$ denote the symmetric regularized self-loop adjacency matrix. We use L_s stacking graph convolution layers to propagate node-level information in the currency relationship graph $\mathcal{G}$:

$$\mathbf{S}_t^{(0)} = \mathbf{H}_t, \quad \mathbf{S}_t^{(\ell+1)} = \sigma(\hat{\mathbf{A}} \mathbf{S}_t^{(\ell)} \mathbf{W}^{(\ell)} + \mathbf{b}^{(\ell)}), \quad (5)$$

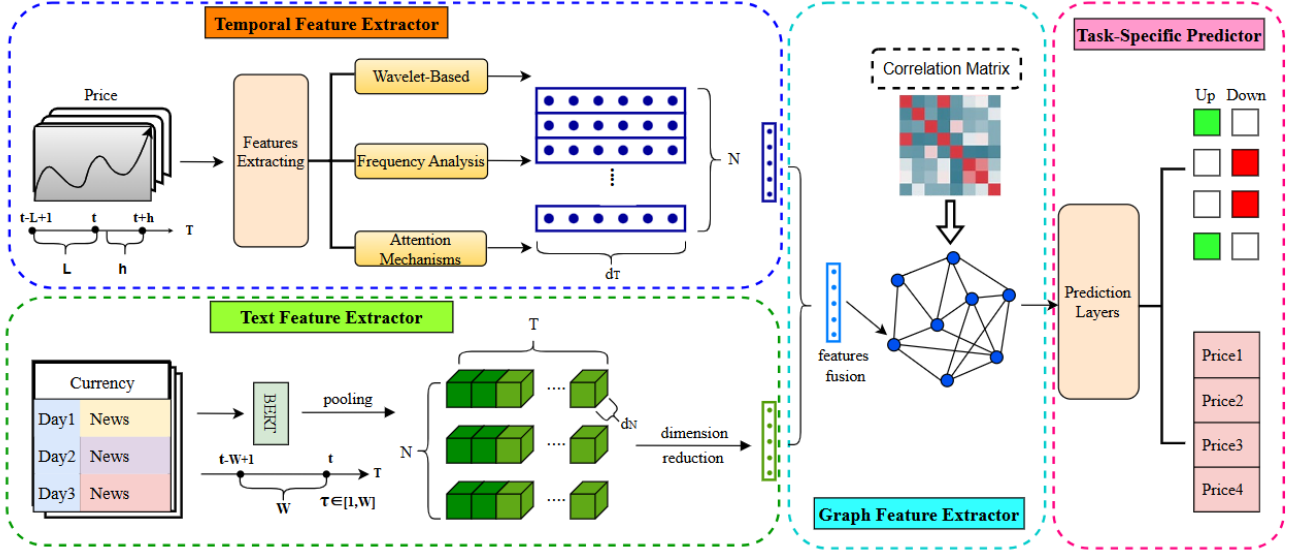


Fig. 1: The Multimodal Multi-Currency Forecasting Framework (MMPF).

where $\ell = 0, \dots, L_s - 1$ and we finally obtain the graph-enhanced representation $\mathbf{S}_t = \mathbf{S}_t^{(L_s)} \in \mathbb{R}^{N \times d_s}$. This process allows the fused temporal-textual features to enrich the spatial dependencies between currencies via GCN.

5) Task-specific predictor:

Based on the overall feature $\mathbf{S}_{i,t}$ for currency i , which lies in the i -th row in $\mathbf{S}_t$, we will consider our downstream tasks. That is, for the regression task, we have:

$$\hat{y}_{i,t+1:t+h}^{\text{reg}} = f_{\text{reg}}(\mathbf{S}_{i,t}), \quad (6)$$

while for the classification task, we let

$$\mathbf{p}_{i,t+1:t+h} = \text{softmax}(\mathbf{W}_c \mathbf{S}_{i,t} + \mathbf{b}_c), \quad (7)$$

$$y_{i,t+1:t+h}^{\text{cls}} \in \{1, \dots, C\}^h, \quad (8)$$

where $\mathbf{W}_c \in \mathbb{R}^{C \times d_s}$ and $\mathbf{b}_c \in \mathbb{R}^C$ are learnable parameters.

C. Training Objectives and Framework

The proposed framework is then formulated as a non-linear mapping:

$$\hat{\mathbf{Y}}_{t+1:t+h} = f_{\theta}(\mathbf{H}_t, \mathcal{G}), \quad (9)$$

where $\mathbf{H}_t$ represents the fused features at time t (based on the fusion of multimodal data before time t), and $\mathcal{G}$ represents the graph of the relationship between cryptocurrencies. The training objective is formulated as a task-conditioned loss with regularization:

$$\mathcal{L}(\theta_1, \theta_2) = \mathbb{E}_{(t,i) \sim \mathcal{D}} \mathcal{L}_{\theta_1}(\hat{y}_{i,t+h}, y_{i,t+h}) + \Omega(\theta_2), \quad (10)$$

where the task-specific loss is defined as:

$$\mathcal{L}_{\theta_1}(\hat{y}_{i,t+h}, y_{i,t+h}) = \begin{cases} (\hat{y}_{i,t+h}^{\text{reg}} - y_{i,t+h}^{\text{reg}})^2, & \text{regression} \\ -\log \mathbf{p}_{i,t+h}[y_{i,t+h}^{\text{cls}}], & \text{classification} \end{cases}$$

Here, (θ_1, θ_2) denotes all model parameters including temporal/text encoders, fusion modules, graph layers, and task-specific heads, $\Omega(\cdot)$ is the regularization term (e.g., weight decay, graph smoothness, temporal consistency).

III. EXPERIMENTS

A. Datasets and Experimental Settings

To validate the performance of the proposed framework, we selected the top six cryptocurrencies by market capitalization on CoinMarketCap as of August 31, 2025: BTC, ETH, XRP, BNB, SOL, and DOGE, together with two relatively lower-ranked mainstream cryptocurrencies: LTC and AVAX. The experimental data come from the OKX exchange and include hourly candlestick closing prices from 11:00:00 on December 21, 2022 to 12:00:00 on June 1, 2025, totaling 21,422 samples. The dataset is divided into training and test sets in the ratio 80% : 20%. Since all cryptocurrency closing prices are denominated in USDT, we will normalize them before all the processing to avoid significant differences in value scales. In addition to price data, we also collected news data from CoinDesk, a leading cryptocurrency news platform. We considered the previous 10 days ($W=10$) of news data in each prediction round. All experimental data will be open-source upon acceptance of the paper.

We employ state-of-the-art pre-trained time-series forecasting models, namely WPMixer [19], TimeMixer [20], and TimeXer [21], to extract temporal representations from cryptocurrency price series. Their original pre-trained weights are loaded and adapted to the cryptocurrency dataset rather than being trained from random initialization. Instead of directly using their forecasting outputs, we use their intermediate hidden representations as temporal features and project them into a unified feature space for multimodal fusion. For comparison, classical temporal backbones, including LSTM and TCN, are

evaluated under the same feature-extraction setting. For news data, we use BERT-base-uncased, a pre-trained Transformer with 12 layers, 768 hidden dimensions, 12 attention heads, a 3072-dimensional intermediate layer, and a maximum sequence length of 512 tokens. To capture the spatial dependencies between cryptocurrencies, we employed a three-layer Lightweight Enhanced GCN (LE-GCN) with hidden layer dimensions of 128, 64, and 32, respectively. Each layer used the ReLU activation function with a dropout probability of 0.5. We also employed batch normalization and residual connections to ensure efficient information propagation, prevent overfitting, and mitigate the effects of oversmoothing. In our experiments, we used a sliding window of 90 historical points ($L=90$) to predict the next time step ($h=1$) and tested two downstream tasks: **1) Classification task:** Predict the upward or downward trend of cryptocurrency prices. Here, we use accuracy (ACC) to evaluate prediction performance [24], where a higher ACC indicates better predictive performance. **2) Regression task:** Predict the actual future prices of cryptocurrencies. We use relative deviation (RD) to evaluate the aggregate-level deviation between the predicted and true price sequences:

$$RD = \left| 1 - \frac{\sum \hat{y}}{\sum y} \right| \times 100\%, \quad (11)$$

where $\hat{y}$ and y denote the predicted and true price values over the evaluation period, respectively. Unlike point-wise error metrics, RD measures the relative deviation between the total predicted value and the total true value. Therefore, it is mainly used to assess whether the predicted sequence captures the overall scale and aggregate trend of the true sequence. A smaller RD indicates a smaller aggregate deviation. We define our Baseline as using only the raw price series and a single TNN model to extract time features and then perform predictions. Building on this baseline, we will add the news model and the GNN model to verify the prediction performance of different frameworks.

B. The Effect of Different TNN Models

In Table I, we compare different temporal backbones for extracting price-series features. For each backbone, the Baseline uses only raw price series without news or graph features. We then add the News module, the LE-GCN module, or both to evaluate their contributions. Results show that LE-GCN generally improves performance, especially in terms of RD, while the effect of news features varies across models. Overall, WPMixer and TimeMixer achieve the best performance, TimeXer and LSTM show moderate results, and TCN performs the weakest but still benefits from graph modeling. These results demonstrate the importance of modeling price-linkage relationships among cryptocurrencies.

C. The Effect of Different GNN Models

Next, we fix WPMixer as the backbone TNN and evaluate the effect of different GNN models. As shown in Table II, GNN architectures have a noticeable impact on prediction performance when combined with the pre-trained WPMixer. For classification, most GNN variants achieve comparable or slightly better ACC than the baseline, with LE-GCN reaching

TABLE I: The Effect of Different Temporal Backbones and Feature Modules

Methods	WPMixer		TimeMixer	
	ACC(%)	RD(%)	ACC(%)	RD(%)
Baseline	51.59%	1.48%	52.67%	0.14%
+News	52.27%	0.08%	52.91%	0.09%
+LE-GCN	52.45%	0.01%	53.73%	0.03%
+LE-GCN and News	52.72%	0.17%	54.01%	0.05%
	TimeXer		LSTM	
	ACC(%)	RD(%)	ACC(%)	RD(%)
Baseline	52.13%	2.04%	51.88%	1.05%
+News	52.23%	2.18%	51.63%	1.82%
+LE-GCN	52.43%	0.88%	52.59%	0.90%
+LE-GCN and News	53.05%	0.32%	52.70%	0.45%
	TCN			
	ACC(%)	RD(%)		
Baseline	49.89%	1.26%		
+News	50.28%	1.42%		
+LE-GCN	50.01%	0.90%		
+LE-GCN and News	52.22%	1.22%		

TABLE II: The Effect of Different GNN Models (with the WPMixer)

configuration	performance	
	classification ACC(%)	regression RD(%)
Baseline	51.59%	1.48%
Basic GCN	51.34%	13.22%
LE-GCN	52.45%	0.01%
GELU-GCN	52.02%	0.71%
GAT	52.16%	0.05%

the highest ACC of 52.45%. For regression, the difference is more significant: Basic GCN increases the RD from 1.48% to 13.22%, suggesting that naive graph convolution may cause over-smoothing or information loss. In contrast, improved GNNs greatly reduce the error, with LE-GCN achieving the best RD of 0.01%, followed by GAT with 0.05%. Overall, pre-trained temporal features combined with a well-designed GNN, especially LE-GCN or GAT, lead to more robust prediction performance.

IV. CONCLUSION

This study leverages price time series, news information, and inter-cryptocurrency relationships to predict cryptocurrency prices and trends. TNNs extract temporal features, while GNNs capture structural dependencies among cryptocurrencies. Experimental results show that temporal encoders initialized with original pre-trained weights generally outperform classical models trained from scratch, demonstrating the benefit of pre-trained time-series representations. Multimodal feature integration further improves trend prediction and relationship modeling, although the effectiveness of news features depends heavily on data quality, relevance, and preprocessing. Future work will explore LLM-based methods to better represent and integrate price and news features.

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